Solutions to "My Best Mathematical and Logic Puzzles" by Martin Gardner

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1. Yes, on any point of a circle centered on the South Pole with radius r miles satisfying:

$$r = 1 + \frac{1}{2c\pi}$$

for $c \in \mathbb{N}$ (the natural numbers 1, 2, 3...)

ex. $r = 1 + \frac{1}{2\pi}$, $1 + \frac{1}{4\pi}$, $1 + \frac{1}{6\pi}$ ...

Because you walk 1 mile south and end up on a circle of radius $\frac{1}{2c\pi}$. The circumference of that circle is $2\pi\left(\frac{1}{2c\pi}\right) = \frac{1}{c}$

so, 1, or $\frac{1}{2}$, $\frac{1}{3}$... Walking 1 mile east you will go around the pole c times and end up where you started before walking east, then walk 1 mile north to where you began.